

Milestone-4 Documentation

Incident Type Analysis

1. Introduction

In this milestone, we analyze different types of disasters and understand their patterns. We also connect this analysis with previous milestones like time trends and state-wise analysis.

2. Libraries Used

Pandas

Pandas is used to handle and analyze data.

Use in project:

- Load dataset
 - Clean data
 - Group and count values
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NumPy

NumPy is used for numerical operations.

Use in project:

- Used in charts for creating positions and calculations
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Matplotlib

Matplotlib is used to create basic charts.

- Bar chart
 - Pie chart
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Seaborn

Seaborn is used for better and more attractive charts.

- Heatmap
 - Improved bar charts
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3. Data Preparation

Before analysis, we prepare the data:

- Rename columns (like Disaster Type → incidentType)
 - Convert date into year and month
 - Remove missing values
 - Convert Yes/No into 1/0
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4. Incident Type Analysis

We check which disaster happens most.

How we do it:

- Count each incident type
- Create bar chart and pie chart

Purpose:

- Find most common disasters
 - Understand overall distribution
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5. State vs Incident Analysis

We check which state faces which disaster.

How we do it:

- Group data by state and incident type
- Create heatmap and stacked bar chart

Purpose:

- Identify disaster-prone states

- Understand regional patterns
-

6. Assistance Program Analysis

We analyze government support programs.

Columns used:

- `ihProgramDeclared` → Individual help
- `paProgramDeclared` → Public help

How we do it:

- Group by incident type
- Compare assistance using bar chart

Purpose:

- Find which disasters need more support
 - Compare types of assistance
-

7. Visualization

We use charts to understand data easily.

Charts used:

- Bar chart → comparison
 - Pie chart → percentage
 - Heatmap → intensity
 - Stacked bar → contribution
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8. Combining All Milestones

We combine all previous work:

- Milestone-1 → Data cleaning
- Milestone-2 → Time analysis
- Milestone-3 → State analysis
- Milestone-4 → Incident analysis

Result:

Complete understanding of disasters.

9. Conclusion

This analysis helps us:

- Find most common disasters
- Identify affected states
- Understand assistance programs

It is useful for planning and decision making.

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